

05-earth-day-science

Earth Day Science Investigation — Full Content

Visual Design Notes

- **Color Palette:** Earth tones — forest green, ocean blue, warm brown, sky blue, sunshine yellow
 - **Style:** Nature-inspired with leaf borders, water droplets, recycling symbols, globe graphics
 - **Layout:** Clean lab report style with colorful section headers and data tables
 - **Fonts:** Clean sans-serif, bolded section headers, easy-to-read body text
 - **Each investigation has a unique color accent** (water=blue, decomp=brown, plants=green, waste=orange, greenhouse=red)
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Investigation 1: Water Filtration Challenge

Teacher Background

Students build water filters from common materials and test how well they clean “dirty” water. This connects to real-world water treatment and access to clean water.

Materials (per group of 3-4)

- 2-liter plastic bottle (cut in half)
- Cotton balls, sand, gravel, activated charcoal (optional)
- Coffee filters or cheesecloth
- “Dirty” water (mix water with soil, small leaves, food coloring)
- Clear cups for collecting filtered water

- Ruler, stopwatch

Procedure

1. **Ask a Question:** How can we clean dirty water using natural materials?
2. **Research:** Discuss how water treatment plants clean water. What materials might trap dirt?
3. **Hypothesize:** “I think the best filter will use ____ because ____”
4. **Design:** Sketch your filter. Layer materials inside the inverted bottle top. The order matters!
 - Suggested layers (bottom to top): coffee filter, cotton balls, charcoal, sand, gravel
5. **Test:** Pour 1 cup of dirty water through your filter. Collect the water that comes out.
6. **Observe & Record:**
 - Color of water before: ____
 - Color of water after: ____
 - Clarity (1=very cloudy, 5=very clear) before: ____ after: ____
 - Time to filter through: ____ seconds
 - Smell before vs. after: ____
7. **Compare:** Look at other groups’ results. Which design worked best? Why?
8. **Reflect:** Could you drink this water? (NO! Filtering removes particles but not bacteria. Discuss.)

Data Collection Sheet

Observation	Before Filtering	After Filtering
Color		
Clarity (1-5)		
Visible particles		
Smell		
Filter time	N/A	

Connection Questions

1. Why is clean water important for living things?
2. What materials in your filter worked best? Why do you think so?
3. About 2 billion people worldwide don't have access to clean drinking water. How does this investigation help you understand that problem?

TEKS: Sci 4.1(A-B) Plan and design investigations; Sci 5.9(D) Human impact; Sci 3.1(E) Collect and record data

Investigation 2: Decomposition Derby

Teacher Background

Students predict, then observe over 2-3 weeks, which materials decompose fastest. Great for discussing landfills, composting, and material choices.

Materials

- 6 clear plastic cups or jars with lids (poke small holes)
- Soil (enough to fill each cup)
- Test items: banana peel, newspaper, plastic bag piece, aluminum foil, leaf, Styrofoam piece
- Spray bottle with water
- Labels/markers
- Camera or drawing supplies (to document changes)

Procedure

1. **Set up (Day 1):** Label each cup with the material name. Fill halfway with soil. Place one test item in each cup. Cover with more soil. Mist with water. Cover.
2. **Predict:** Rank the items from “will decompose fastest” to “will decompose slowest.” Explain your reasoning.
3. **Observe (Days 3, 7, 14, 21):** Carefully dig up each item. Observe changes. Draw or photograph. Re-bury.

4. Record Data:

Material	Day 3	Day 7	Day 14	Day 21	Decomposed?
Banana peel					
Newspaper					
Plastic bag					
Aluminum foil					
Leaf					
Styrofoam					

5. **Analyze:** Create a bar graph showing decomposition level (1=no change, 5=completely gone)

6. **Discuss:** Why do some materials decompose and others don't? What does this mean for landfills?

Reflection Questions

1. Which item decomposed the most? The least? Were you surprised?
2. Why is it important to reduce our use of materials that don't decompose?
3. What is composting, and how does it connect to this investigation?
4. Name 3 things your family could do to produce less non-decomposable waste.

TEKS: Sci 3.9(A) Observe and describe environmental changes; Sci 5.9(D) Identify human impact on future resources; Sci 4.1(E) Collect and record data

Investigation 3: Pollution & Plant Growth

Teacher Background

Students grow plants with regular water vs. “polluted” water (vinegar solution) to observe the impact of pollution on living things.

Materials (per group)

- 6 small pots or cups with drainage holes
- Potting soil
- Fast-growing seeds (bean or grass seeds)
- 3 spray bottles labeled: “Clean Water,” “Mild Pollution” (1 tsp vinegar per cup of water), “Heavy Pollution” (2 tbsp vinegar per cup)
- Ruler
- Sunny window or grow light

Procedure

1. **Set up:** Plant 2 seeds in each of 6 pots. Label 2 pots “Clean,” 2 pots “Mild Pollution,” 2 pots “Heavy Pollution.”
2. **Hypothesize:** What will happen to plants watered with polluted water?
3. **Care for plants:** Water each pair with its labeled water every other day (same amount). Place all in same light.
4. **Observe daily for 10-14 days:** Measure plant height, count leaves, note color and health.

Data Table

Day	Clean Plant Height	Mild Pollution Height	Heavy Pollution Height
3			
5			
7			
10			
14			

5. **Graph:** Create a line graph comparing growth of all 3 groups over time.

6. **Conclude:** How did pollution affect plant growth? What does this tell us about real pollution?

Reflection Questions

1. How did the polluted water affect the plants compared to clean water?
2. In real life, what types of pollution can affect plants and water?
3. If pollution affects plants, how might it affect animals and people too?
4. What can communities do to prevent water pollution?

TEKS: Sci 4.9(A) Investigate basic needs of organisms; Sci 5.9(B-C) Predict outcomes of changes in ecosystems; Sci 5.1(B) Design experimental investigations

Investigation 4: Classroom Waste Audit

Teacher Background

Students collect and categorize one day's classroom waste to understand consumption patterns and identify ways to reduce waste. (Wear gloves! Remove any unsafe items beforehand.)

Materials

- Disposable gloves for all students
- Large tarp or trash bags (to spread waste on)
- Scale (kitchen or postal scale)
- Sorting bins labeled: Paper, Plastic, Food Waste, Metal, Other
- Recording sheets
- Calculator

Procedure

1. **Collect:** Save ALL classroom waste for one full school day (teacher pre-screens for safety).
2. **Sort:** Wearing gloves, sort waste into categories on the tarp.

3. Count & Weigh: Count items and weigh each category.

Data Collection

Category	Number of Items	Weight (oz or g)	Percentage of Total
Paper			
Plastic			
Food Waste			
Metal			
Other			
TOTAL			

4. Calculate percentages: $(\text{Category weight} \div \text{Total weight}) \times 100$

5. Create a circle/pie graph showing waste breakdown

6. Analyze:

- What type of waste do we produce the most?
- How much could be recycled instead of thrown away?
- How much could be composted?
- What could we REDUCE (not use at all)?

Action Plan

Students create a “Classroom Waste Reduction Plan” with 3 specific, actionable steps the class can take. Vote on the top idea and implement it!

TEKS: Sci 5.9(D) Human impact on resources; Math 4.9(A) Represent data; Math 5.8(A-C) Data analysis; Sci 3.1(E) Collect and record data

Investigation 5: Greenhouse Effect Model

Teacher Background

Students model the greenhouse effect using sealed vs. open containers to observe heat trapping. Connects to climate science.

Materials (per group)

- 2 identical clear containers (plastic boxes or glass jars)
- 2 thermometers
- Plastic wrap
- Soil (thin layer in each)
- Lamp with incandescent bulb OR sunny window
- Timer
- Recording sheets

Procedure

1. **Set up:** Place thin layer of soil in both containers. Put a thermometer in each. Cover ONE container tightly with plastic wrap (this is the “greenhouse”). Leave the other open.
2. **Record starting temperatures:** Both should be the same.
3. **Place under lamp or in sunlight.**
4. **Record temperatures every 5 minutes for 30 minutes:**

Time	Open Container (°F)	Covered Container (°F)	Difference
Start			
5 min			
10 min			
15 min			
20 min			
25 min			
30 min			

5. **Graph:** Create a double line graph showing both temperatures over time.
6. **Discuss:** The plastic wrap acts like greenhouse gases in Earth's atmosphere — it lets heat in but traps it inside.

Reflection Questions

1. Which container got hotter? By how much?
2. How is the plastic wrap similar to greenhouse gases in Earth's atmosphere?
3. What are some things that add greenhouse gases to our atmosphere?
4. What can people do to reduce greenhouse gas emissions?

TEKS: Sci 5.8(A) Differentiate between weather and climate; Sci 4.1(B) Design investigations; Math 5.8(A-C) Represent and interpret data

Earth Day Pledge Card

My Earth Day Pledge

I, _____, promise to help protect our planet by:

- ☐ _____
- ☐ _____
- ☐ _____

One thing I learned today: _____

Signed: _____ **Date:** _____

Conservation Poster Project

Create a poster about ONE of these topics: - Water conservation - Reducing, reusing, recycling - Protecting animal habitats - Air quality and pollution - Saving energy

Your poster must include: - A catchy title/slogan - At least 3 facts - At least 1 graphic/illustration - A call to action (what should people DO?)

Rubric: | Category | 4 (Excellent) | 3 (Good) | 2 (Developing) | 1 (Beginning) | |—|—|—|
 —|—| | Content/Facts | 4+ accurate facts | 3 facts | 2 facts | 1 or inaccurate | | Visual
 Appeal | Colorful, neat, creative | Colorful and neat | Some color/effort | Minimal effort | |
 Call to Action | Clear, specific, inspiring | Clear and specific | Present but vague | Missing |
 | TEKS Connection | Strong science connection | Good connection | Weak connection |
 No connection |

Google Slides Version Outline

Slide 1: Title — Earth/globe graphic, green/blue theme

Slides 2-6: Investigation overviews (one per investigation)

Slides 7-11: Digital data collection sheets (editable tables)

Slides 12-16: Reflection question pages

Slide 17: Earth Day pledge (fillable)

Slide 18: Vocabulary cards

Slide 19: “Earth Hero” certificate